Junsung Kim

wnstjd123@gm.gist.ac.kr github.com/ClementKim linkedin.com/in/ClementKim

Research Interests

Vision–language learning and visual reasoning; interaction-conditioned mixture-of-experts; hyperspectral imaging for face anti-spoofing and identification.

Education

Gwangju Institute of Science and Technology (GIST) 2025 – Present
M.S. in Electrical Engineering and Computer Science Gwangju, South Korea
Expected graduation: February 2027

Ajou University 2019 – 2025
B.S. in Software and Computer Science Suwon, South Korea
B.A. in Political Science and Diplomacy

Research Experience

Graduate Researcher, INFONET Lab, GIST 2025 – Present
Advisor: Prof. Heung-No Lee Gwangju, South Korea

IP-MoE: Vision–Language Modeling

- Proposed and implemented a vision–language connector using bilinear token–patch interactions for expert weighting and language-model conditioning; integrated frozen CLIP encoders, a dense mixture of six spatial experts, and Llama-3.2-3B-Instruct.
- Built preprocessing and two-stage distributed training pipelines for LLaVA-NeXT dataset, using LoRA and BF16 across **six NVIDIA A100 GPUs**.
- Tested whether bilinear image–text interactions and learned expert weighting improve visual perception and reasoning, comparing against additive interactions and uniform weighting across **seven benchmarks**.
- Analyzed routing entropy, subtask performance, and prediction patterns, identifying nearly uniform late-training gates and answer-option imbalance on CV-Bench-3D.

Hyperspectral Face Anti-Spoofing and Identification

- Extended a VGG-style CNN with four sparsely gated experts for hyperspectral face anti-spoofing and identity recognition.
- Implemented preprocessing, training, and evaluation on a **4,374 images from 54 participants**; compared performance with a CNN-only baseline.

Beyond Answer Preservation: Private LLM Reasoning

- Contributed to dataset construction for a scientific-reasoning study whose benchmark comprises **160 workflows and 800 tasks across four domains**.
- Provided feedback on the method and algorithm, and reviewed the manuscript's reasoning and logical flow for scientific evidence minimization.

Technical Skills

Programming and frameworks: Python, PyTorch
Tools and environment: Ubuntu/Linux, Docker
Languages: Korean, English

Publications and Manuscripts

Manuscripts Under Review

[1] **Junsung Kim**, Youngin Choi, Minjun Song, Sumin Kim, Surin Lee, and Heung-No Lee, "IP-MoE: Studying spatial token–patch interactions and dense expert weighting," *Submitted to ICLR 2027*, under review.

[2] Surin Lee, **Junsung Kim**, Minjun Song, Sumin Kim, and Heung-No Lee, "Beyond answer preservation: Scientific evidence minimization for private LLM reasoning," *Submitted to ICLR 2027*, under review.

Accepted Workshop Paper

[3] Minjun Song, Sumin Kim, **Junsung Kim**, Yebeen Lee, and Heung-No Lee, "Retrieval-augmented generation for symbolic folk-art interpretation: Comparing dense and graph-based retrieval using Korean Minhwa," *ORACLE Workshop at EMNLP*, 2026, accepted.

Published Domestic Conference Papers

[4] **Junsung Kim**, Youngin Choi, and Heung-No Lee, "An HSI-based CNN-MoE framework for face anti-spoofing and identification," *KICS Winter Conference*, Feb. 2026, pp. 204–205.

[5] Minjun Song, Sumin Kim, **Junsung Kim**, and Heung-No Lee, "AI-assisted dataset creation pipeline for Korean Minhwa," *KICS Winter Conference*, Feb. 2026, pp. 1596–1597.

[6] Yongtae Lee, **Junsung Kim**, and Heung-No Lee, "A prompt-based pipeline for generative game asset synthesis and evaluation," *KICS Winter Conference*, Feb. 2026, pp. 1604–1605.

Honors and Awards

Korean Government Scholarship 2025 – Present
Full tuition and stipend for graduate study.

The Academic Frontier 2025 Apr. 2025
AI Agent & Blockchain Innovation Challenge (idea competition) — **2nd Place**
Prepared and delivered a presentation to competition judges on **Planisphere**, a proposed decentralized platform for AI-agent prediction competitions.

Leadership and Community Service

GIST PIUM — Knowledge-Sharing Volunteer Mar. 2025 – Present
Student volunteer program offering science education and academic and career mentoring to school students, including those in underserved communities.

AI Nights, Ajou University — Organizer & Speaker Sep. 2024
Organized an on-campus AI conference and presented on K-means and DBSCAN.

Daesin-Learning, Ajou University — Team Leader Jul. – Sep. 2024
Led a university-supported machine-learning and deep-learning study group; coordinated schedules, budget, and learning materials.

National Service

National Election Commission — Social Service Agent Feb. 2021 – Nov. 2022
Completed mandatory service as an administrative assistant.